

Unit 11. The Singular Value Decomposition

1 Singular systems and matrix factors

1.1 Full and compact decompositions

Definition 11.1 (Singular-value decomposition). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $m, n \geq 1$, and $q = \min(m, n)$. A **singular-value decomposition** of $\mathbf{A}$ is

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}',$$

where $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$ are orthogonal and

$$\Sigma_{ij} = \begin{cases} \sigma_i, & i = j \leq q, \\ 0, & \text{otherwise,} \end{cases} \quad \sigma_1 \geq \cdots \geq \sigma_q \geq 0.$$

The σ_i are the **singular values**; write $\sigma_i(\mathbf{A})$ when the matrix must be displayed. If exactly r are positive, write

$$\mathbf{U} = (\mathbf{U}_r, \mathbf{U}_0), \quad \mathbf{V} = (\mathbf{V}_r, \mathbf{V}_0), \quad \mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_r).$$

Then

$$\mathbf{A} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r' = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i'$$

is the **compact SVD**; the empty sum is $\mathbf{O}$ when $r = 0$.

Origin: A&M Ex. 8.38; H §21.12(a), eqs. (21.12.4)–(21.12.6); S Thm. 11.4

Theorem 11.2 (Existence and singular-value uniqueness). Every $\mathbf{A} \in \mathbb{R}^{m \times n}$ has an SVD. Moreover,

$$\sigma_i^2 = \lambda_i(\mathbf{A}'\mathbf{A}), \quad 1 \leq i \leq q,$$

where the eigenvalues of $\mathbf{A}'\mathbf{A}$ are in decreasing order. Hence the ordered singular values and

$$r = \text{rank}(\mathbf{A}) = \#\{i : \sigma_i > 0\}$$

are unique; the singular vectors need not be unique.

Origin: A&M Ex. 8.38; H Thm. 21.12.1, Cor. 21.12.2, and Thm. 21.12.3; S Thm. 11.4

Proof. The matrix $\mathbf{A}'\mathbf{A}$ is symmetric positive semidefinite. Its spectral decomposition gives an orthonormal basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ and

$$\mathbf{A}'\mathbf{A}\mathbf{v}_i = \lambda_i\mathbf{v}_i, \quad \lambda_1 \geq \dots \geq \lambda_n \geq 0.$$

Since

$$\text{null}(\mathbf{A}'\mathbf{A}) = \text{null}(\mathbf{A}), \quad \text{rank}(\mathbf{A}'\mathbf{A}) = \text{rank}(\mathbf{A}) = r,$$

exactly r of the λ_i are positive. Also $r \leq q$; set $\sigma_i = \sqrt{\lambda_i}$ for $1 \leq i \leq q$. For $1 \leq i \leq r$, define

$$\mathbf{u}_i = \frac{\mathbf{A}\mathbf{v}_i}{\sigma_i}.$$

Then

$$\mathbf{u}_i' \mathbf{u}_j = \frac{\mathbf{v}_i' \mathbf{A}' \mathbf{A} \mathbf{v}_j}{\sigma_i \sigma_j} = \frac{\lambda_j \mathbf{v}_i' \mathbf{v}_j}{\sigma_i \sigma_j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Extend $\mathbf{u}_1, \dots, \mathbf{u}_r$ to an orthonormal basis of $\mathbb{R}^m$ and form $\mathbf{U}$; form $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$. For $j > r$,

$$\|\mathbf{A}\mathbf{v}_j\|^2 = \mathbf{v}_j' \mathbf{A}' \mathbf{A} \mathbf{v}_j = 0,$$

so $\mathbf{A}\mathbf{v}_j = \mathbf{0}$. Therefore

$$\mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma}, \quad \mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}'.$$

For every SVD,

$$\mathbf{A}'\mathbf{A} = \mathbf{V}\mathbf{\Sigma}'\mathbf{\Sigma}\mathbf{V}',$$

so the ordered eigenvalues of $\mathbf{A}'\mathbf{A}$ determine the σ_i . If a positive eigenvalue is repeated, its orthonormal eigenbasis may be rotated; bases for either null space may also be changed. $\square$

Corollary 11.3 (Paired singular-vector equations). For the compact SVD $\mathbf{A} = \mathbf{U}_r\mathbf{D}\mathbf{V}_r'$,

$$\mathbf{A}\mathbf{V}_r = \mathbf{U}_r\mathbf{D}, \quad \mathbf{A}'\mathbf{U}_r = \mathbf{V}_r\mathbf{D},$$

and hence

$$\mathbf{A}'\mathbf{A}\mathbf{V}_r = \mathbf{V}_r\mathbf{D}^2, \quad \mathbf{A}\mathbf{A}'\mathbf{U}_r = \mathbf{U}_r\mathbf{D}^2.$$

Once either member of a positive singular pair is chosen, the other is fixed by

$$\mathbf{u}_i = \sigma_i^{-1}\mathbf{A}\mathbf{v}_i, \quad \mathbf{v}_i = \sigma_i^{-1}\mathbf{A}'\mathbf{u}_i.$$

Origin: A&M Ex. 8.39; H Thm. 21.12.3, eqs. (21.12.10)–(21.12.13)

Proof. The first identity follows from $\mathbf{V}_r' \mathbf{V}_r = \mathbf{I}_r$. Transposition gives $\mathbf{A}' = \mathbf{V}_r \mathbf{D} \mathbf{U}_r'$ and the second identity. Multiplication by $\mathbf{A}'$ and $\mathbf{A}$, respectively, gives the two symmetric eigenproblems. Taking column i in the first two identities gives the pairing formulas. $\square$

1.2 The four fundamental subspaces

Theorem 11.4 (Four subspaces from one SVD). Let $\mathbf{A} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r'$ have rank r , with $\mathbf{U} = (\mathbf{U}_r, \mathbf{U}_0)$ and $\mathbf{V} = (\mathbf{V}_r, \mathbf{V}_0)$. Then

$$\begin{aligned} \text{col}(\mathbf{A}) &= \text{col}(\mathbf{U}_r), & \text{null}(\mathbf{A}') &= \text{col}(\mathbf{U}_0), \\ \text{row}(\mathbf{A}) &= \text{col}(\mathbf{V}_r), & \text{null}(\mathbf{A}) &= \text{col}(\mathbf{V}_0). \end{aligned}$$

Thus $\mathbf{U}$ and $\mathbf{V}$ give orthonormal bases for all four fundamental subspaces. Empty-column matrices represent the zero subspace.

Origin: H Ex. 21.28; S Thm. 11.4, eqs. (11.4)–(11.5)

Proof. From $\mathbf{A} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r'$,

$$\text{col}(\mathbf{A}) \subseteq \text{col}(\mathbf{U}_r), \quad \mathbf{A} \mathbf{V}_r \mathbf{D}^{-1} = \mathbf{U}_r,$$

so $\text{col}(\mathbf{A}) = \text{col}(\mathbf{U}_r)$. Similarly, $\text{row}(\mathbf{A}) = \text{col}(\mathbf{V}_r)$. For $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{Ax} = \mathbf{0} \iff \mathbf{DV}'_r \mathbf{x} = \mathbf{0} \iff \mathbf{V}'_r \mathbf{x} = \mathbf{0} \iff \mathbf{x} \in \text{col}(\mathbf{V}_0).$$

Applying the same argument to $\mathbf{A}' = \mathbf{V}_r \mathbf{D} \mathbf{U}'_r$ gives $\text{null}(\mathbf{A}') = \text{col}(\mathbf{U}_0)$. □

Example 11.1 (A differencing matrix). For

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix},$$

set

$$\begin{aligned} \mathbf{u}_1 &= 2^{-1/2}(1, -1)', & \mathbf{v}_1 &= 6^{-1/2}(1, -2, 1)', \\ \mathbf{u}_2 &= 2^{-1/2}(1, 1)', & \mathbf{v}_2 &= 2^{-1/2}(1, 0, -1)', \\ & & \mathbf{v}_0 &= 3^{-1/2}(1, 1, 1)'. \end{aligned}$$

Direct multiplication gives

$$\mathbf{Av}_1 = \sqrt{3} \mathbf{u}_1, \quad \mathbf{Av}_2 = \mathbf{u}_2, \quad \mathbf{Av}_0 = \mathbf{0},$$

and therefore

$$\mathbf{A} = \sqrt{3} \mathbf{u}_1 \mathbf{v}_1' + \mathbf{u}_2 \mathbf{v}_2', \quad \text{null}(\mathbf{A}) = \text{span}(\mathbf{v}_0).$$

Origin: Course formulation

1.3 Partial isometries and positive-semidefinite factors

Definition 11.5 (Partial isometry). A matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$ is a **partial isometry** with initial space $\mathcal{S} \subseteq \mathbb{R}^n$ and final space $\mathcal{T} \subseteq \mathbb{R}^m$ if

$$\mathbf{W}\mathbf{x} = \mathbf{0} \quad (\mathbf{x} \in \mathcal{S}^\perp), \quad (\mathbf{W}\mathbf{x})'(\mathbf{W}\mathbf{y}) = \mathbf{x}'\mathbf{y} \quad (\mathbf{x}, \mathbf{y} \in \mathcal{S}), \quad \mathbf{W}\mathcal{S} = \mathcal{T}.$$

Origin: Course formulation from H Ex. 21.27

Definition 11.6 (Canonical rectangular polar decompositions). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, a pair $(\mathbf{W}, \mathbf{H})$ is its **canonical right polar decomposition** if

$$\mathbf{A} = \mathbf{W}\mathbf{H}, \quad \mathbf{H} = (\mathbf{A}'\mathbf{A})^{1/2},$$

and $\mathbf{W}$ is a partial isometry with initial space $\text{row}(\mathbf{A})$ and final space $\text{col}(\mathbf{A})$. A pair $(\mathbf{K}, \mathbf{W})$ is its

canonical left polar decomposition if

$$\mathbf{A} = \mathbf{KW}, \quad \mathbf{K} = (\mathbf{AA}')^{1/2},$$

with the same requirements on $\mathbf{W}$.

Origin: Course formulation from A&M Ex. 8.40; H Thm. 21.12.3; S Thm. 10.1

Theorem 11.7 (Canonical rectangular polar decomposition). Let $\mathbf{A} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r'$ and define

$$\mathbf{W} = \mathbf{U}_r \mathbf{V}_r', \quad \mathbf{H} = \mathbf{V}_r \mathbf{D} \mathbf{V}_r', \quad \mathbf{K} = \mathbf{U}_r \mathbf{D} \mathbf{U}_r'.$$

Then

$$\mathbf{H} = (\mathbf{A}'\mathbf{A})^{1/2}, \quad \mathbf{K} = (\mathbf{AA}')^{1/2}, \quad \mathbf{A} = \mathbf{WH} = \mathbf{KW}.$$

Thus $(\mathbf{W}, \mathbf{H})$ and $(\mathbf{K}, \mathbf{W})$ are the canonical right and left polar decompositions of $\mathbf{A}$. The matrix $\mathbf{W}$ is the unique partial isometry with initial space $\text{row}(\mathbf{A})$, final space $\text{col}(\mathbf{A})$, and $\mathbf{A} = \mathbf{WH}$. Moreover,

$$\mathbf{W}'\mathbf{W} = \mathbf{V}_r \mathbf{V}_r', \quad \mathbf{WW}' = \mathbf{U}_r \mathbf{U}_r'.$$

If $\mathbf{A}$ has full column rank, $\mathbf{W}'\mathbf{W} = \mathbf{I}_n$; only when $m = n = r$ is $\mathbf{W}$ orthogonal.

Origin: Adapted from A&M Ex. 8.40; H Thm. 21.12.3; S Thm. 10.1

Proof. The paired singular-vector equations give

$$\mathbf{A}'\mathbf{A} = \mathbf{V}_r\mathbf{D}^2\mathbf{V}_r', \quad \mathbf{A}\mathbf{A}' = \mathbf{U}_r\mathbf{D}^2\mathbf{U}_r'.$$

Both $\mathbf{H}$ and $\mathbf{K}$ are positive semidefinite and square to the displayed matrices; uniqueness of the positive semidefinite square root identifies them. Also,

$$\mathbf{W}\mathbf{H} = \mathbf{U}_r\mathbf{V}_r'\mathbf{V}_r\mathbf{D}\mathbf{V}_r' = \mathbf{A}, \quad \mathbf{K}\mathbf{W} = \mathbf{U}_r\mathbf{D}\mathbf{U}_r'\mathbf{U}_r\mathbf{V}_r' = \mathbf{A}.$$

For $\mathbf{x} = \mathbf{V}_r\mathbf{c} \in \text{row}(\mathbf{A})$ and $\mathbf{y} = \mathbf{V}_r\mathbf{d} \in \text{row}(\mathbf{A})$,

$$(\mathbf{W}\mathbf{x})'(\mathbf{W}\mathbf{y}) = (\mathbf{U}_r\mathbf{c})'(\mathbf{U}_r\mathbf{d}) = \mathbf{c}'\mathbf{d} = \mathbf{x}'\mathbf{y}.$$

Moreover, $\mathbf{W}\mathbf{V}_0 = \mathbf{O}$ and $\mathbf{W} \text{row}(\mathbf{A}) = \text{col}(\mathbf{U}_r) = \text{col}(\mathbf{A})$, so $\mathbf{W}$ has the stated partial isometry. If $\mathbf{Z}$ has the same properties and $\mathbf{A} = \mathbf{Z}\mathbf{H}$, then

$$\sigma_i\mathbf{Z}\mathbf{v}_i = \mathbf{Z}\mathbf{H}\mathbf{v}_i = \mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i \quad (1 \leq i \leq r), \quad \mathbf{Z}\mathbf{V}_0 = \mathbf{O}.$$

Thus $\mathbf{ZV} = \mathbf{WV}$ and $\mathbf{Z} = \mathbf{W}$. Finally,

$$\mathbf{W}'\mathbf{W} = \mathbf{V}_r \mathbf{V}_r', \quad \mathbf{W}\mathbf{W}' = \mathbf{U}_r \mathbf{U}_r'.$$

The rank statements give the last assertions. □

2 Norms, approximation, and rank thresholds

2.1 Spectral and Frobenius norms

Definition 11.8 (Two matrix norms). For $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{m \times n}$, define

$$\|\mathbf{A}\|_2 = \max_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|, \quad \|\mathbf{A}\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)^{1/2} = \{\text{tr}(\mathbf{A}'\mathbf{A})\}^{1/2}.$$

The first is the **spectral norm**; the second is the **Frobenius norm**.

Origin: H eqs. (6.1.5) and (6.1.9); S Def. 7.4 and §7.1.4

Theorem 11.9 (Norms from singular values). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have singular values $\sigma_1 \geq \dots \geq \sigma_q \geq 0$. Then

$$\|\mathbf{A}\|_2 = \sigma_1, \quad \|\mathbf{A}\|_F = \left(\sum_{i=1}^q \sigma_i^2 \right)^{1/2}.$$

If $\mathbf{P}$ and $\mathbf{Q}$ are conformable orthogonal matrices, then

$$\|\mathbf{PAQ}\|_2 = \|\mathbf{A}\|_2, \quad \|\mathbf{PAQ}\|_F = \|\mathbf{A}\|_F.$$

For conformable $\mathbf{A}, \mathbf{B}$,

$$\|\mathbf{AB}\|_2 \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_2, \quad \|\mathbf{AB}\|_F \leq \min\{\|\mathbf{A}\|_2 \|\mathbf{B}\|_F, \|\mathbf{A}\|_F \|\mathbf{B}\|_2\}.$$

Origin: H Lemma 8.4.2 and eq. (21.12.14); S §7.1.4

Proof. Write $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i$ with $\sum_i c_i^2 = 1$. Then

$$\|\mathbf{Ax}\|^2 = \sum_{i=1}^q \sigma_i^2 c_i^2 \leq \sigma_1^2, \quad \|\mathbf{Av}_1\| = \sigma_1.$$

Thus $\|\mathbf{A}\|_2 = \sigma_1$. Cyclicity of trace gives

$$\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{V}\mathbf{\Sigma}'\mathbf{\Sigma}\mathbf{V}') = \text{tr}(\mathbf{\Sigma}'\mathbf{\Sigma}) = \sum_{i=1}^q \sigma_i^2.$$

If $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}'$, then

$$\mathbf{PAQ} = (\mathbf{PU})\mathbf{\Sigma}(\mathbf{Q}'\mathbf{V})',$$

an SVD with the same singular values. This proves both invariances. The definition of the spectral norm gives

$$\|\mathbf{ABx}\| \leq \|\mathbf{A}\|_2 \|\mathbf{Bx}\| \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_2 \|\mathbf{x}\|.$$

If $\beta_1, \dots, \beta_s$ are the columns of $\mathbf{B}$, then

$$\|\mathbf{AB}\|_F^2 = \sum_{j=1}^s \|\mathbf{A}\beta_j\|^2 \leq \|\mathbf{A}\|_2^2 \sum_{j=1}^s \|\beta_j\|^2 = \|\mathbf{A}\|_2^2 \|\mathbf{B}\|_F^2.$$

Apply this inequality to $\mathbf{B}'\mathbf{A}'$ and use $\|\mathbf{C}'\|_2 = \|\mathbf{C}\|_2$ and $\|\mathbf{C}'\|_F = \|\mathbf{C}\|_F$, which follow from the transposed SVD. $\square$

Corollary 11.10 (Rank-one norms). For $\mathbf{a} \in \mathbb{R}^m$ and $\beta \in \mathbb{R}^n$,

$$\|\mathbf{a}\beta'\|_2 = \|\mathbf{a}\beta'\|_F = \|\mathbf{a}\| \|\beta\|.$$

Proof. The assertion is zero if either vector is zero. Otherwise,

$$\mathbf{a}\boldsymbol{\beta}' = \frac{\mathbf{a}}{\|\mathbf{a}\|}(\|\mathbf{a}\|\|\boldsymbol{\beta}\|)\frac{\boldsymbol{\beta}'}{\|\boldsymbol{\beta}\|}$$

is a compact SVD with one singular value. □

Definition 11.11 (Symmetric dilation). For $\mathbf{M} \in \mathbb{R}^{m \times n}$, its **symmetric dilation** is

$$\mathcal{D}(\mathbf{M}) = \begin{pmatrix} \mathbf{0} & \mathbf{M} \\ \mathbf{M}' & \mathbf{0} \end{pmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}.$$

Origin: Course formulation from S Thm. 6.3 and Prop. 6.2

2.2 Eckart–Young–Mirsky approximation

Definition 11.12 (Truncated SVD). For an SVD $\mathbf{A} = \sum_{i=1}^q \sigma_i \mathbf{u}_i \mathbf{v}_i'$ and $0 \leq k \leq q$, define

$$\mathbf{A}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i', \quad \mathbf{A}_0 = \mathbf{O}, \quad \sigma_{q+1} = 0.$$

Then $\text{rank}(\mathbf{A}_k) \leq k$ and $\mathbf{A}_q = \mathbf{A}$.

Origin: H §21.12(c), preceding Thm. 21.12.4; MC §15

Theorem 11.13 (Eckart–Young–Mirsky). For $0 \leq k \leq q$,

$$\min_{\text{rank}(\mathbf{B}) \leq k} \|\mathbf{A} - \mathbf{B}\|_2 = \|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1},$$

and

$$\min_{\text{rank}(\mathbf{B}) \leq k} \|\mathbf{A} - \mathbf{B}\|_F = \|\mathbf{A} - \mathbf{A}_k\|_F = \left(\sum_{i=k+1}^q \sigma_i^2 \right)^{1/2}.$$

The sum is zero when $k = q$.

Origin: Course formulation from H Thm. 21.12.4; MC §15

Proof. For $k = q$, $\mathbf{A}_q = \mathbf{A}$ and both minima equal $0 = \sigma_{q+1}$. Let $k < q$ and $\text{rank}(\mathbf{B}) \leq k$. Since

$$\dim \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{k+1}) = k + 1, \quad \dim \text{null}(\mathbf{B}) \geq n - k,$$

the two subspaces have a common unit vector $\mathbf{x} = \sum_{i=1}^{k+1} c_i \mathbf{v}_i$. Hence

$$\|\mathbf{A} - \mathbf{B}\|_2^2 \geq \|(\mathbf{A} - \mathbf{B})\mathbf{x}\|^2 = \|\mathbf{A}\mathbf{x}\|^2 = \sum_{i=1}^{k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2.$$

For $k \geq 1$, choose $\mathbf{C} \in \mathbb{R}^{m \times k}$ with $\mathbf{C}'\mathbf{C} = \mathbf{I}_k$ and $\text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{C})$. Then $\mathbf{B} = \mathbf{C}\mathbf{L}$ for $\mathbf{L} = \mathbf{C}'\mathbf{B}$, and

$$\mathbf{C}'(\mathbf{I} - \mathbf{C}\mathbf{C}') = \mathbf{0}.$$

Therefore

$$\begin{aligned} \|\mathbf{A} - \mathbf{C}\mathbf{L}\|_F^2 &= \|(\mathbf{I} - \mathbf{C}\mathbf{C}')\mathbf{A}\|_F^2 + \|\mathbf{C}'\mathbf{A} - \mathbf{L}\|_F^2 \\ &\geq \|\mathbf{A}\|_F^2 - \text{tr}(\mathbf{C}'\mathbf{A}\mathbf{A}'\mathbf{C}). \end{aligned}$$

The Ky Fan trace principle, applied to $\mathbf{A}\mathbf{A}'$, gives

$$\mathrm{tr}(\mathbf{C}'\mathbf{A}\mathbf{A}'\mathbf{C}) \leq \sum_{i=1}^k \sigma_i^2.$$

Thus

$$\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \sum_{i=k+1}^q \sigma_i^2.$$

For $k = 0$ the same inequality is $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$. Finally, $\mathbf{A} - \mathbf{A}_k$ has singular values $\sigma_{k+1}, \dots, \sigma_q$ and zeros, so $\mathbf{A}_k$ attains both bounds. $\square$

2.3 Condition ratios and numerical rank

Definition 11.14 (Condition ratio and condition number). For a nonzero matrix $\mathbf{A}$ of rank r , define its nonzero singular-value ratio by

$$\gamma_2(\mathbf{A}) = \frac{\sigma_1}{\sigma_r}.$$

Only for a square nonsingular matrix is the **spectral-norm condition number** defined here:

$$\kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2.$$

Origin: Course formulation from S §7.1.4 and p. 200

Definition 11.15 (Numerical rank at a threshold). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, set $q = \min(m, n)$. For an absolute threshold $\tau \geq 0$, define

$$\text{rank}_\tau(\mathbf{A}) = \#\{i \in \{1, \dots, q\} : \sigma_i > \tau\}.$$

For $\mathbf{A} \neq \mathbf{O}$, a relative threshold $\varepsilon \geq 0$ means $\tau = \varepsilon \sigma_1$.

Origin: Course formulation

Theorem 11.16 (Condition identities and right-hand-side sensitivity). If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is nonsingular, then

$$\kappa_2(\mathbf{A}) = \gamma_2(\mathbf{A}) = \frac{\sigma_1}{\sigma_n}.$$

For $\mathbf{Ax} = \boldsymbol{\beta} \neq \mathbf{0}$ and $\mathbf{A} \delta \mathbf{x} = \delta \boldsymbol{\beta}$,

$$\frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa_2(\mathbf{A}) \frac{\|\delta \boldsymbol{\beta}\|}{\|\boldsymbol{\beta}\|},$$

and equality is attainable. If a rectangular $\mathbf{A}$ has full column rank, then

$$\kappa_2(\mathbf{A}'\mathbf{A}) = \gamma_2(\mathbf{A})^2;$$

if it has full row rank, then

$$\kappa_2(\mathbf{AA}') = \gamma_2(\mathbf{A})^2.$$

Origin: Course formulation from S §7.1.4 and p. 200

Proof. The transposed SVD gives

$$\mathbf{A}^{-1} = \mathbf{V} \operatorname{diag}(\sigma_1^{-1}, \dots, \sigma_n^{-1}) \mathbf{U}', \quad \|\mathbf{A}^{-1}\|_2 = \sigma_n^{-1}.$$

Moreover,

$$\|\delta \mathbf{x}\| \leq \|\mathbf{A}^{-1}\|_2 \|\delta \boldsymbol{\beta}\|, \quad \|\boldsymbol{\beta}\| = \|\mathbf{Ax}\| \leq \|\mathbf{A}\|_2 \|\mathbf{x}\|.$$

Multiplication gives the bound. With

$$\mathbf{x} = \mathbf{v}_1, \quad \delta\boldsymbol{\beta} = t\mathbf{u}_n \quad (t \neq 0),$$

one has $\boldsymbol{\beta} = \sigma_1 \mathbf{u}_1$ and $\delta\mathbf{x} = t\sigma_n^{-1} \mathbf{v}_n$, so equality holds. For full column rank, the eigenvalues of $\mathbf{A}'\mathbf{A}$ are $\sigma_1^2, \dots, \sigma_n^2$; the first identity follows. The full-row case follows from the nonzero eigenvalues of $\mathbf{A}\mathbf{A}'$. $\square$

Theorem 11.17 (Distance to lower rank). If $\mathbf{A} \neq \mathbf{O}$ has rank r , then

$$\min_{\text{rank}(\mathbf{B}) < r} \|\mathbf{A} - \mathbf{B}\|_2 = \sigma_r, \quad \frac{\sigma_r}{\|\mathbf{A}\|_2} = \frac{1}{\gamma_2(\mathbf{A})}.$$

For every $\tau \geq 0$,

$$\text{rank}_\tau(\mathbf{A}) = \min\{k \in \{0, \dots, q\} : \|\mathbf{A} - \mathbf{A}_k\|_2 \leq \tau\}.$$

Origin: Course formulation from H Thm. 21.12.4

Proof. Set $k = r - 1$ in Theorem 11.13; then $\sigma_{k+1} = \sigma_r$. Division by $\|\mathbf{A}\|_2 = \sigma_1$ gives the relative identity. Also,

$$\|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1} \leq \tau \iff \#\{i : \sigma_i > \tau\} \leq k.$$

The least such k is $\text{rank}_\tau(\mathbf{A})$. □

3 Moore–Penrose inverse and coefficient penalties

3.1 The four Penrose equations

Definition 11.18 (Moore–Penrose inverse). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, a matrix $\mathbf{G} \in \mathbb{R}^{n \times m}$ is a **Moore–Penrose inverse** of $\mathbf{A}$ if

$$\mathbf{AGA} = \mathbf{A}, \quad \mathbf{GAG} = \mathbf{G}, \quad (\mathbf{AG})' = \mathbf{AG}, \quad (\mathbf{GA})' = \mathbf{GA}.$$

These are the four **Penrose equations**.

Origin: A&M §10.3, p. 185; H Thm. 20.1.1; S Thm. 11.5

Theorem 11.19 (Existence, uniqueness, and SVD formula). Every real matrix has exactly one Moore–Penrose inverse. For

$$\mathbf{A} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r',$$

it is

$$\mathbf{A}^+ = \mathbf{V}_r \mathbf{D}^{-1} \mathbf{U}_r'.$$

For $\mathbf{A} = \mathbf{O}$, this is the conformable null matrix.

Origin: A&M Exs. 10.23–10.24; H Thm. 20.1.1 and §21.12(b); S Thm. 11.5

Proof. Let $\mathbf{G}_0 = \mathbf{V}_r \mathbf{D}^{-1} \mathbf{U}_r'$. Then

$$\mathbf{A} \mathbf{G}_0 = \mathbf{U}_r \mathbf{U}_r', \quad \mathbf{G}_0 \mathbf{A} = \mathbf{V}_r \mathbf{V}_r',$$

so both products are symmetric. Moreover,

$$\mathbf{A} \mathbf{G}_0 \mathbf{A} = \mathbf{A}, \quad \mathbf{G}_0 \mathbf{A} \mathbf{G}_0 = \mathbf{G}_0.$$

Thus $\mathbf{G}_0$ satisfies all four equations.

For uniqueness, let $\mathbf{G}$ satisfy them. If $r = 0$, then $\mathbf{G} = \mathbf{G} \mathbf{A} \mathbf{G} = \mathbf{O}$. Suppose $r \geq 1$, use a full SVD $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}'$, and set $\mathbf{Y} = \mathbf{V}' \mathbf{G} \mathbf{U}$. Partition

$$\mathbf{\Sigma} = \begin{pmatrix} \mathbf{D} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} \mathbf{Y}_{11} & \mathbf{Y}_{12} \\ \mathbf{Y}_{21} & \mathbf{Y}_{22} \end{pmatrix},$$

where $\mathbf{Y}_{11} \in \mathbb{R}^{r \times r}$. Orthogonal multiplication transforms the Penrose equations into

$$\mathbf{\Sigma} \mathbf{Y} \mathbf{\Sigma} = \mathbf{\Sigma}, \quad \mathbf{Y} \mathbf{\Sigma} \mathbf{Y} = \mathbf{Y}, \quad (\mathbf{\Sigma} \mathbf{Y})' = \mathbf{\Sigma} \mathbf{Y}, \quad (\mathbf{Y} \mathbf{\Sigma})' = \mathbf{Y} \mathbf{\Sigma}.$$

The last two equations give

$$\mathbf{D}\mathbf{Y}_{12} = \mathbf{0}, \quad \mathbf{Y}_{21}\mathbf{D} = \mathbf{0},$$

hence $\mathbf{Y}_{12} = \mathbf{0}$ and $\mathbf{Y}_{21} = \mathbf{0}$. The first equation gives

$$\mathbf{D}\mathbf{Y}_{11}\mathbf{D} = \mathbf{D} \implies \mathbf{Y}_{11} = \mathbf{D}^{-1}.$$

The second then gives $\mathbf{Y}_{22} = \mathbf{0}$. Therefore

$$\mathbf{G} = \mathbf{V}\mathbf{Y}\mathbf{U}' = \mathbf{V}_r\mathbf{D}^{-1}\mathbf{U}'_r = \mathbf{G}_0.$$

□

Corollary 11.20 (The two canonical projectors). For every $\mathbf{A} \in \mathbb{R}^{m \times n}$,

$$\mathbf{A}\mathbf{A}^+ = \mathbf{U}_r\mathbf{U}'_r, \quad \mathbf{A}^+\mathbf{A} = \mathbf{V}_r\mathbf{V}'_r.$$

Thus $\mathbf{A}\mathbf{A}^+$ is the orthogonal projector onto $\text{col}(\mathbf{A})$ and $\mathbf{A}^+\mathbf{A}$ is the orthogonal projector onto

$\text{row}(\mathbf{A})$. Also,

$\mathbf{I}_m - \mathbf{A}\mathbf{A}^+$ projects onto $\text{null}(\mathbf{A}')$, $\mathbf{I}_n - \mathbf{A}^+\mathbf{A}$ projects onto $\text{null}(\mathbf{A})$.

Origin: A&M Ex. 10.26; H Thm. 20.5.1 and Cor. 20.5.2; S §11.5, text following Prop. 11.5

Proof. The SVD formula gives the first two identities. The four-subspace theorem identifies their column spaces. Since

$$\mathbf{I}_m = \mathbf{U}_r \mathbf{U}_r' + \mathbf{U}_0 \mathbf{U}_0', \quad \mathbf{I}_n = \mathbf{V}_r \mathbf{V}_r' + \mathbf{V}_0 \mathbf{V}_0',$$

the complementary-projector assertions follow. □

Corollary 11.21 (Full-rank formulas). If $\mathbf{A}$ is square and nonsingular, $\mathbf{A}^+ = \mathbf{A}^{-1}$. If $\mathbf{A}$ has full column rank, then

$$\mathbf{A}^+ = (\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'.$$

If $\mathbf{A}$ has full row rank, then

$$\mathbf{A}^+ = \mathbf{A}'(\mathbf{A}\mathbf{A}')^{-1}.$$

Origin: A&M Ex. 10.31; H Eqs. (20.2.1)–(20.2.2); S Thm. 11.5 and p. 177

Proof. For full column rank, $r = n$, $\mathbf{V}_r = \mathbf{V}$, and

$$(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}' = (\mathbf{V}\mathbf{D}^2\mathbf{V}')^{-1}\mathbf{V}\mathbf{D}\mathbf{U}'_r = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}'_r = \mathbf{A}^+.$$

The full-row formula follows by transposition. In the square nonsingular case either formula satisfies both inverse equations. $\square$

3.2 Minimum-norm least squares

Theorem 11.22 (All least-squares solutions and the shortest one). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{\beta} \in \mathbb{R}^m$. The complete set of minimisers of $\|\boldsymbol{\beta} - \mathbf{A}\mathbf{x}\|$ is

$$\{\mathbf{A}^+\boldsymbol{\beta} + (\mathbf{I}_n - \mathbf{A}^+\mathbf{A})\mathbf{z} : \mathbf{z} \in \mathbb{R}^n\}.$$

Every minimiser has fitted vector and residual

$$\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{A}^+\boldsymbol{\beta}, \quad \mathbf{r} = (\mathbf{I}_m - \mathbf{A}\mathbf{A}^+)\boldsymbol{\beta}.$$

Among the minimisers, $\mathbf{A}^+\boldsymbol{\beta}$ is the unique vector of minimum norm.

Origin: A&M Exs. 10.46–10.47; H Thm. 20.6.1; S Prop. 11.6

Proof. Set $\mathbf{P} = \mathbf{A}\mathbf{A}^+$. Since $(\mathbf{I}_m - \mathbf{P})\boldsymbol{\beta} \in \text{null}(\mathbf{A}')$ and $\mathbf{P}\boldsymbol{\beta} - \mathbf{A}\mathbf{x} \in \text{col}(\mathbf{A})$,

$$\|\boldsymbol{\beta} - \mathbf{A}\mathbf{x}\|^2 = \|(\mathbf{I}_m - \mathbf{P})\boldsymbol{\beta}\|^2 + \|\mathbf{P}\boldsymbol{\beta} - \mathbf{A}\mathbf{x}\|^2.$$

Thus $\mathbf{x}$ minimises the residual norm exactly when

$$\mathbf{A}\mathbf{x} = \mathbf{P}\boldsymbol{\beta} = \mathbf{A}\mathbf{A}^+\boldsymbol{\beta}.$$

For every $\mathbf{z}$,

$$\mathbf{A}\{\mathbf{A}^+\boldsymbol{\beta} + (\mathbf{I}_n - \mathbf{A}^+\mathbf{A})\mathbf{z}\} = \mathbf{P}\boldsymbol{\beta}.$$

Conversely, $\mathbf{A}\mathbf{x} = \mathbf{P}\boldsymbol{\beta}$ implies $\mathbf{x} - \mathbf{A}^+\boldsymbol{\beta} \in \text{null}(\mathbf{A})$, and every $\mathbf{w} \in \text{null}(\mathbf{A})$ satisfies

$$\mathbf{w} = (\mathbf{I}_n - \mathbf{A}^+\mathbf{A})\mathbf{w}.$$

This proves the parametrisation. Finally,

$$\mathbf{A}^+\boldsymbol{\beta} \in \text{row}(\mathbf{A}), \quad (\mathbf{I}_n - \mathbf{A}^+\mathbf{A})\mathbf{z} \in \text{null}(\mathbf{A}),$$

so

$$\|\mathbf{A}^+\boldsymbol{\beta} + (\mathbf{I}_n - \mathbf{A}^+\mathbf{A})\mathbf{z}\|^2 = \|\mathbf{A}^+\boldsymbol{\beta}\|^2 + \|(\mathbf{I}_n - \mathbf{A}^+\mathbf{A})\mathbf{z}\|^2.$$

Equality with the first term forces the null-space component to vanish. □

3.3 Centered data and coefficient penalties

Theorem 11.23 (SVD coordinates of centered data). Let $\mathbf{X} \in \mathbb{R}^{m \times p}$, $m \geq 2$, and

$$\mathbf{C} = \mathbf{I}_m - \frac{1}{m}\mathbf{1}\mathbf{1}', \quad \mathbf{Z} = \mathbf{C}\mathbf{X}.$$

For a compact SVD $\mathbf{Z} = \mathbf{U}_r\mathbf{D}\mathbf{V}_r'$,

$$\frac{1}{m-1}\mathbf{Z}'\mathbf{Z} = \mathbf{V}_r\frac{\mathbf{D}^2}{m-1}\mathbf{V}_r', \quad \mathbf{Z}\mathbf{V}_r = \mathbf{U}_r\mathbf{D}, \quad \mathbf{1}'\mathbf{U}_r = \mathbf{0}'.$$

Thus the right singular vectors are sample principal-component loading directions, the score columns are $\mathbf{U}_r \mathbf{D}$, and their sample variances are $\sigma_i^2/(m-1)$. For $0 \leq k \leq r$, with the empty sum interpreted as $\mathbf{O}$,

$$\mathbf{Z}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i', \quad \|\mathbf{Z} - \mathbf{Z}_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2.$$

Origin: Course formulation from H Thm. 21.12.3 and Eq. (21.12.14)

Proof. The first two identities follow by multiplying the compact SVD. Since $\mathbf{1}'\mathbf{C} = \mathbf{0}'$, one has

$$\mathbf{0}' = \mathbf{1}'\mathbf{Z}\mathbf{V}_r = \mathbf{1}'\mathbf{U}_r\mathbf{D} \implies \mathbf{1}'\mathbf{U}_r = \mathbf{0}'.$$

The sample-variance identities follow from the diagonal entries of $\mathbf{D}^2/(m-1)$. The last identity is the Frobenius part of Theorem 11.13. $\square$

Definition 11.24 (Ridge criterion and estimator). For $\mathbf{X} \in \mathbb{R}^{m \times p}$, $\mathbf{y} \in \mathbb{R}^m$, and $\lambda > 0$, the function $f_\lambda(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2$ is the **ridge criterion**; any minimiser is a **ridge estimator**.

Origin: Course formulation from H Thm. 20.7.1

Theorem 11.25 (Ridge as a singular-value filter). Let $\mathbf{X} \in \mathbb{R}^{m \times p}$ have compact SVD $\mathbf{X} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r'$, let $\mathbf{y} \in \mathbb{R}^m$, and let $\lambda > 0$. The ridge criterion in Definition 11.24 has the unique minimiser

$$\widehat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}'\mathbf{y} = \sum_{i=1}^r \frac{\sigma_i}{\sigma_i^2 + \lambda} \mathbf{v}_i \mathbf{u}_i' \mathbf{y}.$$

Its fitted vector is

$$\mathbf{X}\widehat{\boldsymbol{\beta}}_\lambda = \sum_{i=1}^r \frac{\sigma_i^2}{\sigma_i^2 + \lambda} \mathbf{u}_i \mathbf{u}_i' \mathbf{y}.$$

As $\lambda \downarrow 0$,

$$\widehat{\boldsymbol{\beta}}_\lambda \longrightarrow \mathbf{X}^+ \mathbf{y}, \quad \mathbf{X}\widehat{\boldsymbol{\beta}}_\lambda \longrightarrow \mathbf{X}\mathbf{X}^+ \mathbf{y}.$$

Empty sums are zero when $\mathbf{X} = \mathbf{O}$.

Origin: Course formulation from H Thm. 20.7.1

Proof. The matrix $\mathbf{M} = \mathbf{X}'\mathbf{X} + \lambda\mathbf{I}_p$ is positive definite. Set $\widehat{\boldsymbol{\beta}}_\lambda = \mathbf{M}^{-1}\mathbf{X}'\mathbf{y}$. For $\mathbf{d} = \boldsymbol{\beta} - \widehat{\boldsymbol{\beta}}_\lambda$,

$$f_\lambda(\boldsymbol{\beta}) - f_\lambda(\widehat{\boldsymbol{\beta}}_\lambda) = \mathbf{d}'\mathbf{M}\mathbf{d},$$

which is positive for $\mathbf{d} \neq \mathbf{0}$. This proves existence and uniqueness without differentiation.

Complete $\mathbf{V}_r$ to $\mathbf{V} = (\mathbf{V}_r, \mathbf{V}_0)$. Then

$$\mathbf{M} = \mathbf{V}_r(\mathbf{D}^2 + \lambda\mathbf{I}_r)\mathbf{V}_r' + \lambda\mathbf{V}_0\mathbf{V}_0', \quad \mathbf{X}'\mathbf{y} = \mathbf{V}_r\mathbf{D}\mathbf{U}_r'\mathbf{y}.$$

Multiplication by $\mathbf{M}^{-1}$ gives the coefficient formula; multiplication by $\mathbf{X}$ gives the fitted formula. Finally,

$$\frac{\sigma_i}{\sigma_i^2 + \lambda} \longrightarrow \frac{1}{\sigma_i}, \quad \frac{\sigma_i^2}{\sigma_i^2 + \lambda} \longrightarrow 1,$$

which gives the two limits from the SVD formulas for $\mathbf{X}^+$ and $\mathbf{X}\mathbf{X}^+$. □

4 Computation: Numerical Checks in R

4.1 SVD, four subspaces, and polar factors (§1)

```

1 A <- rbind(c(1, -1, 0), c(0, 1, -1))
2 s <- svd(A, nu = nrow(A), nv = ncol(A))
3 tol <- 1e-12
4 r <- sum(s$d > tol)
5 Ur <- s$u[, 1:r, drop = FALSE]
6 Vr <- s$v[, 1:r, drop = FALSE]
7 D <- diag(s$d[1:r], r, r)
8 H <- Vr %*% D %*% t(Vr)
9 K <- Ur %*% D %*% t(Ur)
10 W <- Ur %*% t(Vr)
11 c(singular_values = s$d,
12   svd_reconstruction = norm(A - Ur %*% D %*% t(Vr), "F"),
13   right_polar = norm(A - W %*% H, "F"),
14   left_polar = norm(A - K %*% W, "F"),
15   initial_projector = norm(t(W) %*% W - Vr %*% t(Vr), "F"),
16   final_projector = norm(W %*% t(W) - Ur %*% t(Ur), "F"),
17   null_residual = norm(A %*% s$v[, (r + 1):ncol(A), drop = FALSE], "F"))

```

```
## singular_values1    singular_values2 svd_reconstruction
```

```
##          1.732e+00          1.000e+00          7.360e-16
##      right_polar      left_polar  initial_projector
##          1.681e-15          8.249e-16          1.755e-16
##  final_projector      null_residual
##          8.924e-16          1.110e-16
```

The numerical rank uses the singular-value threshold 10^{-12} .

4.2 Norms and rank-one approximation (§2.2)

```
1 A1 <- s$d[1] * outer(s$u[, 1], s$v[, 1])
2 B <- outer(c(1, 0), c(1, 0, 0))
3 rbind(
4   formula = c(spectral = s$d[2],
5               frobenius = sqrt(sum(s$d[2:length(s$d)]^2))),
6   truncated = c(spectral = norm(A - A1, "2"),
7               frobenius = norm(A - A1, "F")),
8   other_rank_one = c(spectral = norm(A - B, "2"),
9                     frobenius = norm(A - B, "F")))
```

```
##          spectral frobenius
## formula          1.000      1.000
```

```
## truncated      1.000      1.000
## other_rank_one 1.618      1.732
```

4.3 Condition ratios and numerical rank (§2.3)

```
1 C <- diag(c(1, 1e-4, 1e-7))
2 dc <- svd(C)$d
3 C2 <- diag(c(1, 1e-4, 0))
4 c(condition_formula = dc[1] / dc[3],
5    condition_package = kappa(C, exact = TRUE),
6    gram_formula = (dc[1] / dc[3])^2,
7    gram_package = kappa(crossprod(C), exact = TRUE),
8    rank_tol_1e_6 = sum(dc > 1e-6),
9    rank_tol_1e_8 = sum(dc > 1e-8),
10   rank_loss_distance = norm(C - C2, "2"))
```

```
## condition_formula condition_package gram_formula
##              1e+07              1e+07              1e+14
##      gram_package rank_tol_1e_6 rank_tol_1e_8
##              1e+14              2e+00              3e+00
## rank_loss_distance
##              1e-07
```

Here `kappa(..., exact = TRUE)` uses the singular values; the two displayed numerical ranks use the stated absolute thresholds.

4.4 Pseudoinverse and minimum-norm fit (§3.2)

```
1 X <- rbind(c(1, 0, 1), c(0, 1, 1), c(1, 1, 2), c(2, -1, 1))
2 y <- c(1, 2, 0, 3)
3 sx <- svd(X)
4 manual_cutoff <- 1e-12
5 ginv_relative_tol <- 1e-12
6 ginv_cutoff <- ginv_relative_tol * sx$d[1]
7 rx <- sum(sx$d > manual_cutoff)
8 Xp <- sx$v[, 1:rx, drop = FALSE] %*%
9   diag(1 / sx$d[1:rx], rx, rx) %*%
10  t(sx$u[, 1:rx, drop = FALSE])
11 Xp_package <- MASS::ginv(X, tol = ginv_relative_tol)
12 c(manual_cutoff = manual_cutoff,
13   ginv_cutoff = ginv_cutoff,
14   manual_rank = rx,
15   ginv_rank = sum(sx$d > ginv_cutoff),
16   package_difference = norm(Xp - Xp_package, "F"),
```

```

17 penrose_1 = norm(X %*% Xp %*% X - X, "F"),
18 penrose_2 = norm(Xp %*% X %*% Xp - Xp, "F"),
19 symmetry_left = norm(t(X %*% Xp) - X %*% Xp, "F"),
20 symmetry_right = norm(t(Xp %*% X) - Xp %*% X, "F"))

```

```

##      manual_cutoff      ginv_cutoff      manual_rank
##      1.000e-12         3.407e-12         2.000e+00
##      ginv_rank package_difference      penrose_1
##      2.000e+00         8.100e-17         1.197e-15
##      penrose_2      symmetry_left      symmetry_right
##      2.287e-16         4.857e-16         4.228e-16

```

```

1 xp <- drop(Xp %*% y)
2 xa <- xp + c(1, 1, -1)
3 rbind(
4   minimum_norm = c(residual = sqrt(sum((y - X %*% xp)^2)),
5                     coefficient = sqrt(sum(xp^2))),
6   other_minimizer = c(residual = sqrt(sum((y - X %*% xa)^2)),
7                       coefficient = sqrt(sum(xa^2))))

```

```
##      residual coefficient
```

```
## minimum_norm      2.413      0.9375
## other_minimizer    2.413      1.9695
```

The manual SVD retains $d_i > 10^{-12}$. The call to `MASS::ginv` uses the relative tolerance 10^{-12} and retains $d_i > 10^{-12}d_1$; the output gives both absolute cutoffs and ranks.

4.5 Centered coordinates and ridge (§3.3)

```
1 Xraw <- rbind(c(1, 2, 0), c(2, 0, 1), c(3, 1, 4),
2               c(4, 3, 2), c(5, 2, 5))
3 Z <- scale(Xraw, center = TRUE, scale = FALSE)
4 sz <- svd(Z)
5 svd_cutoff <- 1e-12
6 rz <- sum(sz$d > svd_cutoff)
7 Uz <- sz$u[, 1:rz, drop = FALSE]
8 Vz <- sz$v[, 1:rz, drop = FALSE]
9 Dz <- diag(sz$d[1:rz], rz, rz)
10 S <- crossprod(Z) / (nrow(Z) - 1)
11 c(covariance_svd =
12   norm(S - Vz %*% (Dz %*% Dz / (nrow(Z) - 1)) %*% t(Vz), "F"),
13   score_identity = norm(Z %*% Vz - Uz %*% Dz, "F"),
```

```
14 score_means = sqrt(sum(colSums(Uz %*% Dz)^2)))
```

```
## covariance_svd score_identity score_means
##      2.279e-14      2.661e-14      7.140e-16
```

```
1 y <- c(1, -1, 2, 0, 3)
2 lambdas <- c(0.1, 1, 10)
3 direct <- apply(lambdas, function(lambda)
4   solve(crossprod(Z) + lambda * diag(ncol(Z)), crossprod(Z, y)))
5 via_svd <- apply(lambdas, function(lambda)
6   drop(Vz %*%
7     diag(sz$d[1:rz] / (sz$d[1:rz]^2 + lambda), rz, rz) %*%
8     crossprod(Uz, y)))
9 ginv_relative_tol <- sqrt(.Machine$double.eps)
10 ginv_cutoff <- ginv_relative_tol * sz$d[1]
11 Zp <- MASS::ginv(Z, tol = ginv_relative_tol)
12 c(max_coefficient_difference = max(abs(direct - via_svd)),
13   small_lambda_to_pseudoinverse =
14     norm(matrix(solve(crossprod(Z) + 1e-8 * diag(ncol(Z)),
15       crossprod(Z, y)) - Zp %*% y,
16       ncol = 1), "F"),
```

```

17 ginv_relative_tol = ginv_relative_tol,
18 ginv_cutoff = ginv_cutoff)

```

```

##      max_coefficient_difference small_lambda_to_pseudoinverse
##      1.610e-14          1.857e-08
##      ginv_relative_tol          ginv_cutoff
##      1.490e-08          7.519e-08

```

```

1  apply(lambdas, function(lambda)
2     sz$d[1:rz]^2 / (sz$d[1:rz]^2 + lambda))

```

```

##      [,1] [,2] [,3]
## [1,] 0.9961 0.9622 0.71801
## [2,] 0.9833 0.8549 0.37076
## [3,] 0.9127 0.5110 0.09462

```

The manual SVD rank uses the absolute threshold 10^{-12} . The default `MASS::ginv` relative tolerance is $\sqrt{\text{.Machine\$double.eps}}$; equivalently, it retains $d_i > \sqrt{\text{.Machine\$double.eps}} d_1$. The output gives this relative tolerance and its absolute cutoff. The last matrix contains the fitted-value filter factors $\sigma_i^2 / (\sigma_i^2 + \lambda)$.

5 Pitfalls

5.1 Solving two eigenproblems independently

Positive singular vectors must satisfy

$$\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad \mathbf{A}'\mathbf{u}_i = \sigma_i\mathbf{v}_i.$$

Independent eigensolvers may choose incompatible signs and, within a repeated eigenspace, incompatible orthogonal rotations. Compute one side and obtain the other from the pairing equation.

5.2 Calling a rectangular polar factor orthogonal

For $\mathbf{W} \in \mathbb{R}^{m \times n}$ in the canonical polar decomposition,

$$\mathbf{W}'\mathbf{W} = \mathbf{P}_{\text{row}(\mathbf{A})}, \quad \mathbf{W}\mathbf{W}' = \mathbf{P}_{\text{col}(\mathbf{A})}.$$

Both products equal identity matrices only when $m = n = \text{rank}(\mathbf{A})$.

5.3 Calling singular values eigenvalues

For

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \text{spec}(\mathbf{A}) = \{0, 0\}, \quad \text{spec}(\mathbf{A}'\mathbf{A}) = \{1, 0\}.$$

Thus the singular values are 1, 0. If $\mathbf{A} = \mathbf{A}'$ instead, then $\mathbf{A}'\mathbf{A} = \mathbf{A}^2$, so its singular values are the absolute values of its eigenvalues, reordered decreasingly.

5.4 Squaring a condition ratio outside its domain

The identity

$$\kappa_2(\mathbf{X}'\mathbf{X}) = \gamma_2(\mathbf{X})^2$$

requires full column rank. If $\mathbf{X}$ is column-rank deficient, then $\mathbf{X}'\mathbf{X}$ is singular and $\kappa_2(\mathbf{X}'\mathbf{X})$ is not defined here.

5.5 Reporting numerical rank without a scale

For $c \neq 0$,

$$\text{rank}_\tau(c\mathbf{A}) = \text{rank}_{\tau/|c|}(\mathbf{A}).$$

An absolute threshold is scale dependent; a relative threshold $\tau = \varepsilon\sigma_1$ is invariant under nonzero scalar rescaling.

5.6 Treating the pseudoinverse as identification

If $\mathbf{X} = (\mathbf{x}, \mathbf{x})$, fitted vectors depend on $\beta_1 + \beta_2$ only. Among all pairs with $\beta_1 + \beta_2 = s$,

$$\beta_1^2 + \beta_2^2 = \frac{s^2}{2} + \frac{(\beta_1 - \beta_2)^2}{2}$$

is minimised at $\beta_1 = \beta_2 = s/2$. The pseudoinverse makes this minimum-norm choice; the separate coefficients remain unidentified by the fit.

6 Exercises

Exercise 11.1 (Rank-one structure). Let $\mathbf{0} \neq \mathbf{a} \in \mathbb{R}^m$, $\mathbf{0} \neq \boldsymbol{\beta} \in \mathbb{R}^n$, and $\mathbf{A} = \mathbf{a}\boldsymbol{\beta}'$. Find a compact SVD of $\mathbf{A}$, compute $\mathbf{A}^+$, and identify $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$.

Origin: A&M Ex. 10.33; S Ex. 11.9

Solution. Set

$$\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|}, \quad \mathbf{v} = \frac{\boldsymbol{\beta}}{\|\boldsymbol{\beta}\|}, \quad \sigma = \|\mathbf{a}\| \|\boldsymbol{\beta}\|.$$

Then

$$\mathbf{A} = \mathbf{u}\sigma\mathbf{v}', \quad \mathbf{A}^+ = \mathbf{v}\sigma^{-1}\mathbf{u}' = \frac{\boldsymbol{\beta}\mathbf{a}'}{(\mathbf{a}'\mathbf{a})(\boldsymbol{\beta}'\boldsymbol{\beta})}.$$

Consequently,

$$\mathbf{A}\mathbf{A}^+ = \frac{\mathbf{a}\mathbf{a}'}{\mathbf{a}'\mathbf{a}}, \quad \mathbf{A}^+\mathbf{A} = \frac{\boldsymbol{\beta}\boldsymbol{\beta}'}{\boldsymbol{\beta}'\boldsymbol{\beta}},$$

the orthogonal projectors onto $\text{span}(\mathbf{a})$ and $\text{span}(\boldsymbol{\beta})$. ■

Exercise 11.2 (SVD of a symmetric matrix). Let

$$\mathbf{A} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}', \quad |\lambda_1| \geq \dots \geq |\lambda_n|,$$

be a spectral decomposition of a real symmetric matrix. Construct an SVD of $\mathbf{A}$ and derive its Moore–Penrose inverse.

Origin: A&M Ex. 10.35; H §21.12(a)–(b), eqs. (21.12.14)–(21.12.15)

Solution. Define

$$\delta_i = \begin{cases} 1, & \lambda_i \geq 0, \\ -1, & \lambda_i < 0, \end{cases} \quad \mathbf{\Delta} = \operatorname{diag}(\delta_1, \dots, \delta_n), \quad \mathbf{\Sigma} = \operatorname{diag}(|\lambda_1|, \dots, |\lambda_n|).$$

Then $\mathbf{Q}\mathbf{\Delta}$ and $\mathbf{Q}$ are orthogonal and

$$\mathbf{A} = (\mathbf{Q}\mathbf{\Delta})\mathbf{\Sigma}\mathbf{Q}'.$$

Thus the singular values are the ordered $|\lambda_i|$. With

$$\lambda_i^+ = \begin{cases} \lambda_i^{-1}, & \lambda_i \neq 0, \\ 0, & \lambda_i = 0, \end{cases}$$

the SVD formula gives

$$\mathbf{A}^+ = \mathbf{Q} \operatorname{diag}(\lambda_1^+, \dots, \lambda_n^+) \mathbf{Q}'.$$



Exercise 11.3 (Transpose and involution). Prove

$$(\mathbf{A}')^+ = (\mathbf{A}^+)', \quad (\mathbf{A}^+)^+ = \mathbf{A}.$$

Identify the nonzero singular values of $\mathbf{A}^+$.

Origin: A&M Ex. 10.25(a)–(b); H Thm. 20.5.3(1),(4) and §21.12(b)

Solution. If $\mathbf{A} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r'$, then

$$\mathbf{A}' = \mathbf{V}_r \mathbf{D} \mathbf{U}_r', \quad (\mathbf{A}')^+ = \mathbf{U}_r \mathbf{D}^{-1} \mathbf{V}_r' = (\mathbf{A}^+)',$$

and

$$\mathbf{A}^+ = \mathbf{V}_r \mathbf{D}^{-1} \mathbf{U}_r' \implies (\mathbf{A}^+)^+ = \mathbf{U}_r \mathbf{D} \mathbf{V}_r' = \mathbf{A}.$$

The nonzero singular values of $\mathbf{A}^+$ are $\sigma_1^{-1}, \dots, \sigma_r^{-1}$; decreasing order reverses this list. ■

Exercise 11.4 (Characterising partial isometries). For $\mathbf{W} \in \mathbb{R}^{m \times n}$, prove that the following are equivalent:

- (a) $\mathbf{W}$ is a partial isometry with initial space $\text{row}(\mathbf{W})$ and final space $\text{col}(\mathbf{W})$;
- (b) $\mathbf{W}'\mathbf{W}$ is an orthogonal projector;
- (c) every singular value of $\mathbf{W}$ is 0 or 1;
- (d) $\mathbf{W}^+ = \mathbf{W}'$.

Include $\mathbf{W} = \mathbf{O}$.

Origin: Adapted from H Ex. 20.2

Solution. Let $\mathbf{W} = \mathbf{U}_r \mathbf{D} \mathbf{V}_r'$. For $\mathbf{x} = \mathbf{V}_r \mathbf{c} \in \text{row}(\mathbf{W})$,

$$\|\mathbf{W}\mathbf{x}\|^2 = \mathbf{c}' \mathbf{D}^2 \mathbf{c}, \quad \|\mathbf{x}\|^2 = \mathbf{c}' \mathbf{c}.$$

Moreover,

$$\mathbf{W} \text{col}(\mathbf{V}_0) = \{\mathbf{0}\}, \quad \mathbf{W} \text{row}(\mathbf{W}) = \text{col}(\mathbf{W}).$$

Hence (a) holds exactly when $\mathbf{D}^2 = \mathbf{I}_r$. Also,

$$\mathbf{W}'\mathbf{W} = \mathbf{V}_r \mathbf{D}^2 \mathbf{V}_r', \quad (\mathbf{W}'\mathbf{W})^2 = \mathbf{V}_r \mathbf{D}^4 \mathbf{V}_r',$$

so (b) is equivalent to $\sigma_i^4 = \sigma_i^2$, hence to (c). Finally,

$$\mathbf{W}^+ = \mathbf{V}_r \mathbf{D}^{-1} \mathbf{U}_r', \quad \mathbf{W}' = \mathbf{V}_r \mathbf{D} \mathbf{U}_r',$$

so (c) is equivalent to (d). For $r = 0$, all four assertions hold with initial and final spaces $\{\mathbf{0}\}$. ■

Exercise 11.5 (Nearest orthogonal matrix). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ have a full SVD $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}'$. Prove that

$$\mathbf{Q}_* = \mathbf{U}\mathbf{V}'$$

minimises $\|\mathbf{A} - \mathbf{Q}\|_F$ over all orthogonal $\mathbf{Q}$, with

$$\min_{\mathbf{Q}'\mathbf{Q}=\mathbf{I}_n} \|\mathbf{A} - \mathbf{Q}\|_F^2 = \sum_{i=1}^n (\sigma_i - 1)^2.$$

Prove uniqueness when $\mathbf{A}$ is nonsingular.

Origin: S Ex. 10.4

Solution. For orthogonal $\mathbf{Q}$, set $\mathbf{R} = \mathbf{U}'\mathbf{Q}\mathbf{V}$. Orthogonal invariance gives

$$\begin{aligned}\|\mathbf{A} - \mathbf{Q}\|_F^2 &= \|\Sigma - \mathbf{R}\|_F^2 \\ &= \sum_{i=1}^n \sigma_i^2 + n - 2 \sum_{i=1}^n \sigma_i r_{ii} \\ &\geq \sum_{i=1}^n \sigma_i^2 + n - 2 \sum_{i=1}^n \sigma_i = \sum_{i=1}^n (\sigma_i - 1)^2,\end{aligned}$$

because $r_{ii} \leq 1$. The choice $\mathbf{R} = \mathbf{I}_n$, equivalently $\mathbf{Q} = \mathbf{U}\mathbf{V}'$, attains the bound. If every $\sigma_i > 0$, equality forces $r_{ii} = 1$ for every i . The unit norm of each row of $\mathbf{R}$ then forces all off-diagonal entries to vanish, so $\mathbf{R} = \mathbf{I}_n$. ■

Exercise 11.6 (A rank-loss certificate). Let $\mathbf{A} \neq \mathbf{O}$ have rank r and compact SVD $\sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i'$. Construct a matrix of rank $r - 1$ at spectral distance σ_r from $\mathbf{A}$. Prove that

$$\|\mathbf{E}\|_2 < \sigma_r \implies \text{rank}(\mathbf{A} + \mathbf{E}) \geq r.$$

Origin: Course formulation from H Thm. 21.12.4

Solution. Set

$$\mathbf{B} = \sum_{i=1}^{r-1} \sigma_i \mathbf{u}_i \mathbf{v}_i'.$$

Then $\text{rank}(\mathbf{B}) = r - 1$ and $\|\mathbf{A} - \mathbf{B}\|_2 = \sigma_r$. Theorem 11.17 shows that no lower-rank matrix is closer. If $\text{rank}(\mathbf{A} + \mathbf{E}) < r$, then

$$\sigma_r \leq \|\mathbf{A} - (\mathbf{A} + \mathbf{E})\|_2 = \|\mathbf{E}\|_2,$$

contradicting the strict inequality. 

Exercise 11.7 (Symmetric dilation and singular-value perturbation). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, set $q = \min(m, n)$. Show that the eigenvalues of the symmetric dilation $\mathcal{D}(\mathbf{A})$ from Definition 11.11, in decreasing order, consist of

$$\sigma_1(\mathbf{A}), \dots, \sigma_q(\mathbf{A}), \quad |m - n| \text{ additional zeros}, \quad -\sigma_q(\mathbf{A}), \dots, -\sigma_1(\mathbf{A}).$$

Deduce, for $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$ and $1 \leq i \leq q$, that

$$|\sigma_i(\mathbf{A}) - \sigma_i(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|_2.$$

Origin: Course formulation from S Thm. 6.3 and Prop. 6.2

Solution. Take a full SVD $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}'$. For $1 \leq i \leq q$,

$$\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad \mathbf{A}'\mathbf{u}_i = \sigma_i\mathbf{v}_i.$$

Hence

$$\mathcal{D}(\mathbf{A}) \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix} = \sigma_i \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix}, \quad \mathcal{D}(\mathbf{A}) \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{u}_i \\ -\mathbf{v}_i \end{pmatrix} = -\sigma_i \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{u}_i \\ -\mathbf{v}_i \end{pmatrix}.$$

These $2q$ vectors are orthonormal, including when $\sigma_i = 0$. If $m > n$, then

$$\begin{pmatrix} \mathbf{u}_j \\ \mathbf{0} \end{pmatrix}, \quad n < j \leq m,$$

are $m - n$ further zero-eigenvectors because $\mathbf{u}_j \in \text{null}(\mathbf{A}')$. If $n > m$, then

$$\begin{pmatrix} \mathbf{0} \\ \mathbf{v}_j \end{pmatrix}, \quad m < j \leq n,$$

are $n - m$ further zero-eigenvectors because $\mathbf{v}_j \in \text{null}(\mathbf{A})$. In each case the displayed vectors form an orthonormal basis of $\mathbb{R}^{m+n}$. Hence the spectral list holds, including $\mathbf{A} = \mathbf{O}$. In particular,

$$\|\mathcal{D}(\mathbf{A}) - \mathcal{D}(\mathbf{B})\|_2 = \|\mathcal{D}(\mathbf{A} - \mathbf{B})\|_2 = \sigma_1(\mathbf{A} - \mathbf{B}) = \|\mathbf{A} - \mathbf{B}\|_2.$$

The first q ordered eigenvalues of $\mathcal{D}(\mathbf{A})$ are $\sigma_i(\mathbf{A})$. The earlier Weyl perturbation bound for symmetric matrices states that, for $\mathbf{M} = \mathbf{M}'$ and $\mathbf{N} = \mathbf{N}'$ of the same order,

$$|\lambda_i(\mathbf{M}) - \lambda_i(\mathbf{N})| \leq \|\mathbf{M} - \mathbf{N}\|_2.$$

Since $\mathcal{D}(\mathbf{A})$ and $\mathcal{D}(\mathbf{B})$ are symmetric of order $m + n$,

$$|\sigma_i(\mathbf{A}) - \sigma_i(\mathbf{B})| \leq \|\mathcal{D}(\mathbf{A}) - \mathcal{D}(\mathbf{B})\|_2 = \|\mathbf{A} - \mathbf{B}\|_2.$$



Exercise 11.8 (Block-diagonal pseudoinverse). For arbitrary real matrices $\mathbf{A}$ and $\mathbf{B}$, prove

$$\begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix}^+ = \begin{pmatrix} \mathbf{A}^+ & \mathbf{O} \\ \mathbf{O} & \mathbf{B}^+ \end{pmatrix}.$$

Origin: A&M Ex. 10.34

Solution. Let $\mathbf{M} = \text{diag}(\mathbf{A}, \mathbf{B})$ and $\mathbf{G} = \text{diag}(\mathbf{A}^+, \mathbf{B}^+)$. Block multiplication gives

$$\mathbf{MGM} = \text{diag}(\mathbf{AA}^+\mathbf{A}, \mathbf{BB}^+\mathbf{B}) = \mathbf{M},$$

$$\mathbf{GMG} = \text{diag}(\mathbf{A}^+\mathbf{AA}^+, \mathbf{B}^+\mathbf{BB}^+) = \mathbf{G}.$$

Also $\mathbf{MG}$ and $\mathbf{GM}$ are block diagonal with symmetric diagonal blocks. Hence $\mathbf{G}$ satisfies the four Penrose equations and equals $\mathbf{M}^+$. ■

Exercise 11.9 (Failure and recovery of the reverse-order law). (a) Give exact pseudoinverses showing that $(\mathbf{AB})^+ = \mathbf{B}^+\mathbf{A}^+$ need not hold, using

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

(b) Let $\mathbf{A} \in \mathbb{R}^{m \times r}$ have full column rank and $\mathbf{B} \in \mathbb{R}^{r \times n}$ have full row rank. Prove

$$(\mathbf{AB})^+ = \mathbf{B}^+\mathbf{A}^+.$$

Origin: A&M Ex. 10.36(b); H Ex. 20.1; S Prop. 11.5 and p. 177

Solution. For part (a), the rank-one formula and the inverse of $\mathbf{B}$ give

$$(\mathbf{AB})^+ = \frac{1}{10} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{A}^+ = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B}^+ = \mathbf{B}^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

Hence

$$\mathbf{B}^+ \mathbf{A}^+ = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \neq (\mathbf{AB})^+.$$

For part (b), set $\mathbf{G} = \mathbf{B}^+ \mathbf{A}^+$. The rank assumptions give

$$\mathbf{A}^+ \mathbf{A} = \mathbf{I}_r, \quad \mathbf{B} \mathbf{B}^+ = \mathbf{I}_r.$$

Therefore

$$\mathbf{ABG} = \mathbf{AA}^+, \quad \mathbf{GAB} = \mathbf{B}^+ \mathbf{B},$$

which are symmetric, and

$$(\mathbf{AB})\mathbf{G}(\mathbf{AB}) = \mathbf{AB}, \quad \mathbf{G}(\mathbf{AB})\mathbf{G} = \mathbf{G}.$$

Thus $\mathbf{G}$ satisfies the Penrose equations for $\mathbf{AB}$. 

Exercise 11.10 (Consistent systems). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{\beta} \in \mathbb{R}^m$, prove

$$\mathbf{Ax} = \boldsymbol{\beta} \text{ is consistent} \iff \mathbf{AA}^+\boldsymbol{\beta} = \boldsymbol{\beta}.$$

When it is consistent, prove that all solutions are

$$\mathbf{x} = \mathbf{A}^+\boldsymbol{\beta} + (\mathbf{I}_n - \mathbf{A}^+\mathbf{A})\mathbf{z}, \quad \mathbf{z} \in \mathbb{R}^n,$$

and identify the unique minimum-norm solution.

Origin: A&M Exs. 10.45–10.46; S Prop. 11.6

Solution. If $\boldsymbol{\beta} = \mathbf{Ax}$, then

$$\mathbf{AA}^+\boldsymbol{\beta} = \mathbf{AA}^+\mathbf{Ax} = \mathbf{Ax} = \boldsymbol{\beta}.$$

Conversely, $\mathbf{AA}^+\boldsymbol{\beta} = \boldsymbol{\beta}$ makes $\mathbf{A}^+\boldsymbol{\beta}$ a solution. Under this condition, exact solutions are precisely the least-squares solutions in Theorem 11.22, giving the displayed parametrisation. The same theorem gives the unique minimum-norm solution $\mathbf{A}^+\boldsymbol{\beta}$. ■

Exercise 11.11 (The equation $\mathbf{AXB} = \mathbf{C}$). Let

$$\mathbf{A} \in \mathbb{R}^{m \times n}, \quad \mathbf{B} \in \mathbb{R}^{p \times q}, \quad \mathbf{C} \in \mathbb{R}^{m \times q}.$$

Prove that $\mathbf{AXB} = \mathbf{C}$, with $\mathbf{X} \in \mathbb{R}^{n \times p}$, is consistent if and only if

$$\mathbf{AA}^+\mathbf{CB}^+\mathbf{B} = \mathbf{C}.$$

When it is consistent, prove that all solutions are

$$\mathbf{X} = \mathbf{A}^+\mathbf{CB}^+ + \mathbf{Z} - \mathbf{A}^+\mathbf{AZBB}^+, \quad \mathbf{Z} \in \mathbb{R}^{n \times p}.$$

Origin: A&M Ex. 10.49; S Ex. 11.13

Solution. If $\mathbf{C} = \mathbf{AXB}$, then

$$\mathbf{AA}^+\mathbf{CB}^+\mathbf{B} = \mathbf{AA}^+\mathbf{AXB}^+\mathbf{B} = \mathbf{AXB} = \mathbf{C}.$$

Conversely, the displayed condition gives

$$\mathbf{A}(\mathbf{A}^+\mathbf{CB}^+)\mathbf{B} = \mathbf{AA}^+\mathbf{CB}^+\mathbf{B} = \mathbf{C}.$$

Under this condition, for every $\mathbf{Z} \in \mathbb{R}^{n \times p}$,

$$\begin{aligned}\mathbf{A}\{\mathbf{A}^+\mathbf{C}\mathbf{B}^+ + \mathbf{Z} - \mathbf{A}^+\mathbf{A}\mathbf{Z}\mathbf{B}\mathbf{B}^+\}\mathbf{B} &= \mathbf{C} + \mathbf{A}\mathbf{Z}\mathbf{B} - \mathbf{A}\mathbf{Z}\mathbf{B} \\ &= \mathbf{C}.\end{aligned}$$

If $\mathbf{A}\mathbf{X}\mathbf{B} = \mathbf{C}$, choosing $\mathbf{Z} = \mathbf{X}$ gives

$$\mathbf{A}^+\mathbf{C}\mathbf{B}^+ + \mathbf{X} - \mathbf{A}^+\mathbf{A}\mathbf{X}\mathbf{B}\mathbf{B}^+ = \mathbf{X}.$$

Hence the displayed family is exact. 

Exercise 11.12 (Equal Moore–Penrose projectors). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, prove

$$\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+\mathbf{A} \iff \text{col}(\mathbf{A}) = \text{row}(\mathbf{A}).$$

Deduce the equality when $\mathbf{A} = \mathbf{A}'$. Give a nonsymmetric singular matrix for which the equality also holds.

Origin: A&M Ex. 10.37; H Ex. 20.8

Solution. The two matrices are the orthogonal projectors onto $\text{col}(\mathbf{A})$ and $\text{row}(\mathbf{A})$, respectively. Orthogonal projectors are equal exactly when their ranges are equal; hence

$$\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+\mathbf{A} \iff \text{col}(\mathbf{A}) = \text{row}(\mathbf{A}).$$

If $\mathbf{A} = \mathbf{A}'$, then

$$\text{row}(\mathbf{A}) = \text{col}(\mathbf{A}') = \text{col}(\mathbf{A}).$$

The converse fails. For

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{A}^+ = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

Corollary 11.21 and Exercise 11.8 give

$$\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+\mathbf{A} = \text{diag}(1, 1, 0), \quad \mathbf{A} \neq \mathbf{A}'.$$



Exercise 11.13 (Orthogonal sums of matrices). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$ satisfy

$$\mathbf{A}'\mathbf{B} = \mathbf{0}, \quad \mathbf{A}\mathbf{B}' = \mathbf{0}.$$

Prove

$$(\mathbf{A} + \mathbf{B})^+ = \mathbf{A}^+ + \mathbf{B}^+.$$

Origin: H Ex. 20.6

Solution. Take compact SVDs

$$\mathbf{A} = \mathbf{U}_A \mathbf{D}_A \mathbf{V}_A', \quad \mathbf{B} = \mathbf{U}_B \mathbf{D}_B \mathbf{V}_B'.$$

Empty blocks are omitted when either matrix is zero. The two hypotheses state, respectively, that the column spaces and row spaces are orthogonal. Thus

$$\mathbf{U}_A' \mathbf{U}_B = \mathbf{0}, \quad \mathbf{V}_A' \mathbf{V}_B = \mathbf{0}.$$

Consequently,

$$\mathbf{A} + \mathbf{B} = (\mathbf{U}_A, \mathbf{U}_B) \begin{pmatrix} \mathbf{D}_A & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_B \end{pmatrix} (\mathbf{V}_A, \mathbf{V}_B)'$$

is a compact SVD up to reordering its positive diagonal entries. Taking reciprocals gives

$$(\mathbf{A} + \mathbf{B})^+ = \mathbf{V}_A \mathbf{D}_A^{-1} \mathbf{U}_A' + \mathbf{V}_B \mathbf{D}_B^{-1} \mathbf{U}_B' = \mathbf{A}^+ + \mathbf{B}^+.$$



Exercise 11.14 (Order reversal for singular pseudoinverses). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ be symmetric and satisfy

$$\mathbf{0} \preceq \mathbf{A} \preceq \mathbf{B}.$$

Prove

$$\mathbf{A}^+ - \mathbf{B}^+ \succeq \mathbf{0} \iff \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B}).$$

Origin: H Ex. 20.13

Solution. Exercise 11.2 gives, for $\mathbf{M} = \mathbf{M}' \succeq \mathbf{O}$,

$$\mathbf{M}^+ \succeq \mathbf{O}, \quad \text{null}(\mathbf{M}^+) = \text{null}(\mathbf{M}).$$

If $\mathbf{x} \in \text{null}(\mathbf{B})$, then

$$0 \leq \mathbf{x}'\mathbf{A}\mathbf{x} \leq \mathbf{x}'\mathbf{B}\mathbf{x} = 0 \quad \implies \quad \mathbf{A}\mathbf{x} = \mathbf{0}.$$

Thus $\text{null}(\mathbf{B}) \subseteq \text{null}(\mathbf{A})$.

Suppose $\mathbf{A}^+ - \mathbf{B}^+ \succeq \mathbf{O}$. For $\mathbf{x} \in \text{null}(\mathbf{A})$,

$$0 \leq \mathbf{x}'(\mathbf{A}^+ - \mathbf{B}^+)\mathbf{x} = -\mathbf{x}'\mathbf{B}^+\mathbf{x} \leq 0,$$

so $\mathbf{B}^+\mathbf{x} = \mathbf{0}$ and $\mathbf{x} \in \text{null}(\mathbf{B})$. Hence

$$\text{null}(\mathbf{A}) = \text{null}(\mathbf{B}) \quad \implies \quad \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B}).$$

Conversely, suppose both ranks equal r . The null-space inclusion and rank-nullity give

$$\text{null}(\mathbf{A}) = \text{null}(\mathbf{B}) = \mathcal{N}.$$

If $r = 0$, then $\mathbf{A} = \mathbf{B} = \mathbf{O}$ and the conclusion holds. Let $r \geq 1$, let $\mathbf{Q}_r$ have orthonormal columns spanning $\mathcal{N}^\perp$, and set

$$\mathbf{A}_r = \mathbf{Q}_r' \mathbf{A} \mathbf{Q}_r, \quad \mathbf{B}_r = \mathbf{Q}_r' \mathbf{B} \mathbf{Q}_r.$$

For $\mathbf{0} \neq \mathbf{c} \in \mathbb{R}^r$, the positive-semidefinite null criterion gives

$$\mathbf{c}' \mathbf{A}_r \mathbf{c} = 0 \implies \mathbf{A} \mathbf{Q}_r \mathbf{c} = \mathbf{0} \implies \mathbf{Q}_r \mathbf{c} \in \mathcal{N},$$

which contradicts $\mathbf{Q}_r \mathbf{c} \in \mathcal{N}^\perp \setminus \{\mathbf{0}\}$. Thus, with $\mathbf{P} = \mathbf{Q}_r \mathbf{Q}_r'$,

$$\mathbf{M}(\mathbf{I}_n - \mathbf{P}) = \mathbf{O} = (\mathbf{I}_n - \mathbf{P})\mathbf{M} \quad (\mathbf{M} = \mathbf{A}, \mathbf{B}), \quad \mathbf{M} = \mathbf{PMP}.$$

Congruence now gives

$$\mathbf{O} \prec \mathbf{A}_r \preceq \mathbf{B}_r, \quad \mathbf{A} = \mathbf{Q}_r \mathbf{A}_r \mathbf{Q}_r', \quad \mathbf{B} = \mathbf{Q}_r \mathbf{B}_r \mathbf{Q}_r'.$$

The inverse-order theorem gives

$$\mathbf{A}_r^{-1} - \mathbf{B}_r^{-1} \succeq \mathbf{O}.$$

For $\mathbf{M} = \mathbf{A}, \mathbf{B}$, write $\mathbf{M} = \mathbf{Q}_r \mathbf{M}_r \mathbf{Q}'_r$ and $\mathbf{G} = \mathbf{Q}_r \mathbf{M}_r^{-1} \mathbf{Q}'_r$. Then

$$\mathbf{MG} = \mathbf{GM} = \mathbf{Q}_r \mathbf{Q}'_r, \quad \mathbf{MGM} = \mathbf{M}, \quad \mathbf{GMG} = \mathbf{G}.$$

The Penrose equations and uniqueness give

$$\mathbf{A}^+ = \mathbf{Q}_r \mathbf{A}_r^{-1} \mathbf{Q}'_r, \quad \mathbf{B}^+ = \mathbf{Q}_r \mathbf{B}_r^{-1} \mathbf{Q}'_r,$$

and therefore

$$\mathbf{A}^+ - \mathbf{B}^+ = \mathbf{Q}_r (\mathbf{A}_r^{-1} - \mathbf{B}_r^{-1}) \mathbf{Q}'_r \succeq \mathbf{0}.$$



Exercise 11.15 (Ridge as augmented least squares). Let $\mathbf{X} \in \mathbb{R}^{m \times p}$, $\mathbf{y} \in \mathbb{R}^m$, and $\lambda > 0$. Define

$$\tilde{\mathbf{X}} = \begin{pmatrix} \mathbf{X} \\ \sqrt{\lambda} \mathbf{I}_p \end{pmatrix}, \quad \tilde{\mathbf{y}} = \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix}.$$

Prove that $\tilde{\mathbf{X}}$ has full column rank and that ordinary least squares for $(\tilde{\mathbf{X}}, \tilde{\mathbf{y}})$ gives

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}'\mathbf{y}.$$

If $\hat{\boldsymbol{\beta}}_0$ is any ordinary least-squares minimiser for $(\mathbf{X}, \mathbf{y})$, prove

$$\|\hat{\boldsymbol{\beta}}_\lambda\| \leq \|\hat{\boldsymbol{\beta}}_0\|.$$

Origin: Course formulation from H Thm. 20.7.1

Solution. For $\boldsymbol{\beta} \neq \mathbf{0}$,

$$\|\tilde{\mathbf{X}}\boldsymbol{\beta}\|^2 = \|\mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2 > 0,$$

so $\tilde{\mathbf{X}}$ has full column rank. Also,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\boldsymbol{\beta}\|^2 = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2,$$

and its normal equations are

$$(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}_p)\boldsymbol{\beta} = \mathbf{X}'\mathbf{y}.$$

Let $R(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ and $R_0 = R(\hat{\boldsymbol{\beta}}_0)$. Ridge optimality and ordinary least-squares optimality give

$$R(\hat{\boldsymbol{\beta}}_\lambda) + \lambda\|\hat{\boldsymbol{\beta}}_\lambda\|^2 \leq R_0 + \lambda\|\hat{\boldsymbol{\beta}}_0\|^2, \quad R(\hat{\boldsymbol{\beta}}_\lambda) \geq R_0.$$

Subtracting yields the asserted norm inequality. 